

UST'-USA KRASNOYARSK, Russian Federation

WMO: 360830

Lat: 52.057N

Lon: 92.139E

Elev: 1893

StdP: 13.72

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.5	-20.4	-29.1	1.2	-22.9	-26.1	1.4	-20.1	14.6	-3.3	11.7	-3.0	3.0	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	83.9	63.8	80.4	62.6	77.2	61.8	66.6	78.8	64.8	76.0	63.5	73.8	5.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	91.3	70.6	61.0	85.9	68.7	59.5	81.5	68.0	32.4	78.7	31.0	75.8	29.9	74.2	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Extreme Max WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 16.6	13.6	11.1	DB	-26.2	90.5	5.0	4.1	-29.8	93.5	-32.7	95.8	-35.5	98.1	-39.2	101.1	
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	-8.4	1.8	24.7	40.9	52.0	63.8	66.4	63.1	51.8	38.2	17.2	-1.1
	DBStd	26.83	7.35	8.76	9.59	7.18	7.62	6.82	4.38	4.82	6.18	7.00	10.68	9.29
	HDD50	7287	1810	1350	786	283	67	2	0	0	54	367	984	1584
	HDD65	11366	2275	1770	1251	723	407	101	35	93	398	830	1434	2049
	CDD50	1581	0	0	0	10	129	417	510	406	107	2	0	0
	CDD65	185	0	0	0	0	4	66	80	34	1	0	0	0
	CDH74	1460	0	0	0	1	113	609	510	219	8	0	0	0
	CDH80	325	0	0	0	0	23	166	102	33	1	0	0	0
Wind	WSAvg	3.4	2.7	3.2	3.3	3.9	4.3	4.1	3.7	3.4	3.2	2.9	3.9	2.7
Precipitation	PrecAvg	15.9	0.5	0.3	0.5	0.8	1.3	2.3	3.3	3.0	1.4	0.9	1.0	0.7
	PrecMax	18.6	1.2	0.6	1.2	1.8	2.4	3.5	5.3	4.7	2.2	2.3	2.0	1.5
	PrecMin	13.0	0.2	0.0	0.1	0.1	0.6	1.0	2.1	1.5	0.6	0.4	0.5	0.2
	PrecStd	1.5	0.2	0.2	0.3	0.4	0.5	0.6	0.8	0.8	0.4	0.4	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	27.2	53.3	70.0	83.4	89.5	87.5	83.6	72.4	58.9	41.6	22.9
		MCWB	11.0	24.0	41.3	50.9	58.0	66.6	65.1	64.8	57.6	48.3	36.6	20.4
	2%	DB	6.5	22.1	47.3	64.1	77.1	84.5	83.2	79.3	67.8	54.1	38.0	18.9
		MCWB	5.3	18.7	37.9	48.4	55.5	63.6	65.1	64.0	57.4	45.3	34.4	17.3
	5%	DB	3.2	17.9	43.3	59.1	71.9	80.6	79.9	75.9	65.0	50.8	34.0	14.0
		MCWB	2.1	15.4	35.6	45.4	53.5	61.8	64.1	62.3	55.7	43.8	30.3	12.6
	10%	DB	0.8	14.4	39.4	54.6	67.1	76.9	76.4	72.6	62.1	47.7	30.4	10.5
		MCWB	-0.1	12.3	33.0	43.2	51.2	60.9	63.1	61.2	53.7	42.2	27.1	9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	24.2	41.6	52.5	60.3	68.2	68.9	68.1	60.9	49.2	37.4	20.6
		MCDB	11.7	27.1	51.7	67.4	79.1	85.9	80.1	78.6	69.0	56.8	40.0	22.1
	2%	WB	5.3	18.7	38.7	48.8	57.0	65.6	66.9	65.7	58.8	46.2	34.6	17.2
		MCDB	6.5	21.8	46.3	61.6	73.1	80.2	78.4	76.3	66.3	52.6	37.9	18.9
	5%	WB	2.3	15.5	36.2	46.4	54.6	63.6	65.2	63.9	56.7	44.4	30.7	12.8
		MCDB	3.1	17.7	42.9	57.9	69.4	77.7	75.8	73.0	63.4	50.1	33.7	14.0
	10%	WB	0.1	12.4	33.2	43.8	52.3	61.9	64.0	62.2	54.6	42.5	27.4	9.4
		MCDB	0.8	14.4	38.6	53.5	65.3	74.2	73.8	70.1	61.1	47.5	30.2	10.5
Mean Daily Temperature Range	5% DB	MDBR	8.5	13.3	18.6	21.0	21.7	19.5	16.0	16.2	15.0	11.2	7.7	7.2
		MCDBR	9.9	15.5	21.4	30.0	31.1	25.6	22.7	22.7	19.0	16.9	9.0	9.5
		MCWBR	9.2	13.1	13.4	16.0	14.2	9.9	9.3	10.4	10.2	10.1	7.1	8.1
	5% WB	MCDBR	10.1	15.1	20.4	27.0	28.2	21.8	18.9	18.4	16.7	15.5	8.3	8.5
		MCWBR	9.4	12.9	12.8	14.9	13.9	9.1	8.9	9.7	10.1	9.7	7.2	7.6
Clear-Sky Solar Irradiance	taub	0.289	0.327	0.384	0.436	0.439	0.426	0.424	0.408	0.368	0.357	0.328	0.294	
	taud	2.041	1.974	1.912	2.029	2.093	2.247	2.292	2.304	2.341	2.219	2.081	2.012	
	Ebn at Noon	213	238	247	252	258	263	261	259	257	229	194	184	
	Edh at Noon	25	36	48	49	48	42	39	36	31	29	24	22	
All-Sky Solar Radiation	RadAvg	395	737	1214	1538	1761	1937	1772	1524	1153	693	376	274	
	RadStd	23	37	56	74	120	151	113	110	100	59	35	20	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page